IMDADULLAH RAJI

Mechanical Engineering, BUET · B.Sc. expected June 2027
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Research Interests

Unsteady fluid dynamics · Dynamical systems · Data-driven modeling · Flow control

Education

Bangladesh University of Engineering and Technology, Dhaka 2022–2027 (Expected)
B.Sc. in Mechanical Engineering · CGPA: 3.58

Relevant coursework: Complex Analysis, Differential Equations, Linear Algebra, Numerical Analysis, Computer Programming, Fluid Mechanics I–II

Notre Dame College, Dhaka 2019–2022
Higher Secondary Certificate, Science

Research Experience

Modeling and Control of Flows Over Rapidly Maneuvering Airfoils 2026–present
Undergraduate Thesis · Supervisor: Dr. Mohammad Ali, Professor, Department of Mechanical Engineering, BUET

- Assessing how accurately reduced-order models capture aerodynamic forces under active actuation, with leading-edge blowing as the control input, and developing improved formulations.
- Generating high-fidelity training data from controlled OpenFOAM simulations. The Reynolds number is kept low deliberately so the flow is fully resolved without turbulence modeling, keeping the data faithful to the governing equations rather than to a closure assumption.

Statistical Modeling of Lagrangian Coherent Structures 2026–present

- Addressing computational bottlenecks in LCS calculations: integrating a full particle ensemble requires Jacobians and singular values at every step, and most of that cost goes to trajectories that carry no information about where the attracting sets form.
- Investigating whether early integration steps predict where attracting sets form, allowing dense Lagrangian sampling to be concentrated in critical flow regions.

Autonomous UAV–UGV Swarm Coordination 2025–2026
Supervisor: Dr. Kazi Arafat Rahman, Associate Professor, Department of Mechanical Engineering, BUET

- Implemented a collaborative, autonomous multi-agent UAV–UGV system using edge-computing hardware (Jetson Orin Nano, Raspberry Pi).

Computational Projects

flowkit — Python library under development for OpenFOAM post-processing, including snapshot loading, interpolation onto Cartesian grids, spatial cropping, and Lagrangian trajectory analysis.

openfoam_cases — Reproducible CFD cases developed for the undergraduate thesis. Each case ships the geometry and mesh generation scripts needed to reproduce it and documents the modeling assumptions it rests on.

Publications

N. A. Tanisha, **I. Raji**, A. Guha, N. Anita. *MuADDIBBS: A Hierarchical Outdoor Swarm System with Aerial Supervision Guiding Local Interaction.* 15th International Conference on Mechanical Engineering (ICME), BUET, 2025. SSRN.

Scientific Outreach & Honors

Bangladesh Olympiad on Astronomy and Astrophysics
Academic Coordinator (2026–present); Academic Member (2021–present)

- Set and evaluate problems for regional and national astrophysics olympiads; organize the national camp and the selection process for Bangladesh's IOAA team.

Honorable Mention, International Olympiad on Astronomy and Astrophysics (IOAA), 2021
Bronze Medal, Open World Astronomy Olympiad (OWAO), 2022

Technical Skills

CFD	OpenFOAM and gmsh; ANSYS Fluent
Scientific Computing	Python (NumPy, SciPy); FEniCS; MATLAB/Simulink
Data & Modeling	PySINDy, PyDMD, scikit-learn; PyTorch
Development	Linux, Git, LaTeX